BioEvo v0.1 - Evaluation Specification

Today's work package | The Genome Release | Baseline evaluation contract

1. Objective

BioEvo evaluates whether a model can reason correctly about biological systems, processes and responses. Version 0.1 establishes a reproducible baseline for future comparisons.

2. Baseline domains

Domain	Code
Apoptosis	APO
Environmental Adaptation	ENV
Evolution	EVO
Homeostasis	HOM
Regeneration	REG

3. Common evaluation-case schema

```
id: BIOEVO-DOMAIN-001
version: "0.1"
domain: domain_name
prompt_scenario: >
  Biological question, experiment, condition, perturbation or scenario.
expected_biological_behaviour: >
  Expected scientifically correct biological outcome or reasoning.
required_concepts:
  - concept_1
  - concept_2
failure_conditions:
  - incorrect biological conclusion
  - incorrect causal/mechanistic reasoning
  - missing critical concept
  - unsupported biological claim
difficulty: easy  # easy | medium | hard
score:
  biological_correctness: 0
  mechanistic_reasoning: 0
  required_concepts: 0
  unsupported_claims: 0
max_score: 100
```

4. Primary scoring metric

Dimension	Maximum
Biological correctness	40
Mechanistic reasoning	30
Required concepts	20
Unsupported claims	10

Total BioEvo Case Score	100

Overall BioEvo Score = arithmetic mean of all evaluated case scores. Domain-level scores should also be reported separately. Proposed v0.1 interpretation: 90-100 Excellent; 75-89 Good; 60-74 Partial; below 60 Fail.

5. Failure taxonomy

BIOLOGICAL_ERROR | CAUSAL_ERROR | MECHANISM_ERROR | MISSING_CONCEPT | UNSUPPORTED_CLAIM | CONTRADICTION | MISINTERPRETATION

6. Worked example

```
id: BIOEVO-APO-001
version: "0.1"
domain: apoptosis
prompt_scenario: >
  A cell experiences severe DNA damage that cannot be repaired.
  Predict the expected cellular response and explain the biological reasoning.
expected_biological_behaviour: >
  Recognise that irreparable cellular damage can activate mechanisms
  leading to programmed cell death rather than continued normal proliferation.
required_concepts:
  - DNA damage
  - cellular damage response
  - apoptosis
  - programmed cell death
failure_conditions:
  - predicts continued normal proliferation without justification
  - describes apoptosis as uncontrolled cell death
  - provides an incorrect causal relationship
  - introduces unsupported biological mechanisms
difficulty: medium
max_score: 100
```

7. Evaluator input format

The evaluator receives the evaluation case, the model response and evaluation metadata.

```
case:
  id: BIOEVO-APO-001
  version: "0.1"
  domain: apoptosis
  difficulty: medium
  prompt_scenario: "..."
  expected_biological_behaviour: "..."
  required_concepts: ["DNA damage", "apoptosis"]
  failure_conditions: ["incorrect causal relationship"]
model_response: "Actual answer generated by model"
evaluation:
  rubric_version: "0.1"
  max_score: 100
```

8. Evaluator output format

```
{
  "case_id": "BIOEVO-APO-001",
  "eval_version": "0.1",
  "domain": "apoptosis",
  "difficulty": "medium",
  "scores": {
    "biological_correctness": 40,
    "mechanistic_reasoning": 25,
    "required_concepts": 20,
    "unsupported_claims": 10
  },
  "total_score": 95,
  "failure_conditions_triggered": [],
  "failure_labels": [],
  "pass": true,
  "evaluator_notes": "Biologically correct response with strong reasoning."
}
```

9. Score validation

biological_correctness: 0-40 | mechanistic_reasoning: 0-30 | required_concepts: 0-20 | unsupported_claims: 0-10. Total score is the sum of these dimensions. Proposed v0.1 pass condition: total_score >= 60.

10. Evaluation flow

BioEvo case -> Model Under Evaluation -> model_response -> BioEvo Evaluator -> structured JSON result -> results.jsonl -> summary.json

11. Recommended repository structure

```
README.md
ROADMAP.md
EVAL_SPEC.md
BioEvo_v0.1_Todays_Work.pdf
cases/
  apoptosis/
  environmental_adaptation/
  evolution/
  homeostasis/
  regeneration/
results/
```

12. Definition of done for today

1. Common eval schema defined. 2. 0-100 BioEvo scoring metric defined. 3. Five v0.1 baseline domains recorded. 4. Failure taxonomy defined. 5. Worked example included. 6. Evaluator input/output contract drafted.

Next work: convert the existing domain tests into structured cases without changing the frozen v0.1 evaluation contract.